

DIVYANSHU SINGH CHAUHAN

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PROFESSIONAL SUMMARY

Controls and Guidance Engineer specializing in nonlinear and optimal control, trajectory optimization, and state estimation for autonomous systems. Experienced in designing constrained optimal trajectories using numerical optimization methods, with hands on implementation in Python, MATLAB, and CasADi. Graduate training in autonomous systems and control, with applied research in autonomous spacecraft docking under uncertainty and belief-space planning.

TECHNICAL SKILLS

- **Control & Estimation:** Nonlinear Control, MPC, State Estimation, EKF, UKF
- **Optimization:** Trajectory Optimization, Constrained Nonlinear Programming, CasADi, IPOPT
- **Programming & Simulation:** Python, MATLAB, Simulink, C++, ROS2
- **Software Engineering:** Git, Docker, Linux, CI/CD, Modular Architecture, FastAPI, Agile/Scrum
- **Other Tools:** Jira, Git, Linux, FastAPI, Splunk Phantom, DO-178C*, Docker, Unity

EXPERIENCE

Graduate Student Research Assistant (Spacecraft Trajectory Optimization)

Ann Arbor, MI

University of Michigan

May 2025 – May 2026

- Designed and implemented a three-phase autonomous spacecraft docking trajectory pipeline under realistic operational and terminal constraints.
- Formulated **uncertain terminal constraints** using Monte Carlo sampling and **covariance-based quaternion** perturbations to improve robustness.
- Developed the trajectory optimization framework in **Python** using **CasADi** and the **IPOPT** solver, with extensibility for belief-space planning and **EKF/UKF**-based state estimation.
- Performed simulation-based validation across varying initial conditions and uncertainty levels to evaluate robustness and convergence behavior.

Python Developer

Bangalore, India

Standard Chartered Bank

October 2021 – May 2024

- Implemented and deployed Python-based SOAR automation workflows using **Splunk Phantom**, reducing incident response time by **40%** and analyst workload by **30%** across production security pipelines.
- Built and maintained **RESTful** microservices using **FastAPI**, integrating enterprise security tooling and co-developing a **supervised phishing detection** model (**83%** accuracy) with responsibility for feature engineering and model evaluation, using **Git** and **Jira-driven Agile** workflows.

Deep Learning Summer Intern

Bangalore, India

HyperVerge Inc.

May 2020 – Aug 2020

- Developed **U-Net** and **GAN-assisted autoencoders** for image tampering localization, achieving **91%** detection accuracy on authentic data and significantly improving localization performance on manipulated samples.
- Designed a custom **Conv2D** layer that improved image manipulation detection, increasing precision rates from **75%** to **90%**, and allowing for more accurate insights in data-driven decision-making processes.

PROJECTS

Robust Cascaded Control for a Flexible Inverted Pendulum

Jan 2026 - April 2026

- Architected a **cascaded PID controller** to stabilize a **6-state flexible inverted pendulum**, achieving precise position tracking and rapid angle stabilization within strict $\pm 10^\circ$ safety limits.
- Mitigated structural resonance ($\omega_n = 20$ rad/s) and satisfied Nyquist stability criteria by designing a 6x inner-loop frequency separation, targeting bandwidth reduction, and implementing a custom **notch filter**.
- Validated system robustness against worst-case scenarios including **100N impulse disturbances** and near-zero structural damping ($\zeta = 0.0002$) and verified absolute stability via left-half plane (LHP) pole migration.

Communication-Aware, Physics-Based and Photo-Realistic simulator for multi-robot systems

Jan 2025 - May 2025

Prof. Vasileios Tzoumas, University of Michigan

- Developed multi-robot simulations for semantic mapping using **ROS2** for middleware in **C++** language in **Unity** Environment.
- Worked on migration of the project from **ROS1** to **ROS2** with the implementation of **SlideSLAM**.

Spacecraft ADCS Design, Integration, and Testing (SPACE 584)	Aug 2025 - Dec 2025
- Developed an Extended Kalman Filter (EKF) in MATLAB for real-time attitude estimation; validated robustness via Monte Carlo simulations and 1σ error-bound analysis. - Designed an autonomous state controller with detumble modes (B-Dot) to stabilize spacecraft from high-tip-off rates (1 rev/s) to pointing stability (< 0.5 deg/s). - Implemented Star Tracker cluster identification and IR Sun Sensor center-of-pixel logic to achieve 1° attitude determination accuracy.	

Botlab and Armlab (BotLab & ArmLab)	Aug 2024 - Nov 2024
- Implemented A* planning and particle-filter SLAM on mBot using IMU and LiDAR sensors. - Developed forward and inverse kinematics for a 5-DOF robotic arm; integrated RealSense-based vision control in Python.	

EDUCATION

University of Michigan, Ann Arbor	Michigan, USA
<i>MSE - Aerospace Engineering (GPA: 3.77/4.0); (Subdomain: Autonomous Systems and Control)</i>	<i>Aug 2024 - May 2026</i>
Indian Institute of Technology (IIT) Guwahati	Guwahati, India
<i>B.Tech - Mechanical Engineering (GPA: 8.27/10.0)</i>	<i>July 2017 - July 2021</i>

PUBLICATIONS

Deep Learning via LSTM Models for COVID-19 Infection Forecasting in India, PLOS One	Link
Neural Anomaly Search on Lunar Reconnaissance Orbiter Camera Images, AbSciCon 2022	Link